

TECHNICAL REPORT

AUTOMATED DETECTION OF GLACIAL LAKES FROM MULTI-TEMPORAL SATELLITE IMAGERY (GLOFEAGLES)

Submitted by

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Introduction:

Rapid glacier retreat driven by climate change has resulted in the widespread formation and expansion of glacial lakes in high-altitude mountain regions across the world. These lakes, particularly those dammed by loose moraines or remnant ice, pose a significant hazard in the form of Glacial Lake Outburst Floods (GLOFs). Accurate identification of glacial lakes is a fundamental prerequisite for reliable GLOF hazard assessment and improved flood prediction.

Dataset Description:

The dataset includes 575 png images of size 512x512 which involve various hindrances associated with glacier lakes such as

- i) Cloud cover
- ii) Terrain shadow
- iii) Snow cover
- iv) Moraine
- v) Debris cover
- vi) Turbidity cover

A set of 60 ground truth segmentation masks were provided (10 from each class of hindrance) from the 575 images. A sample from each class is given below at Fig 1.

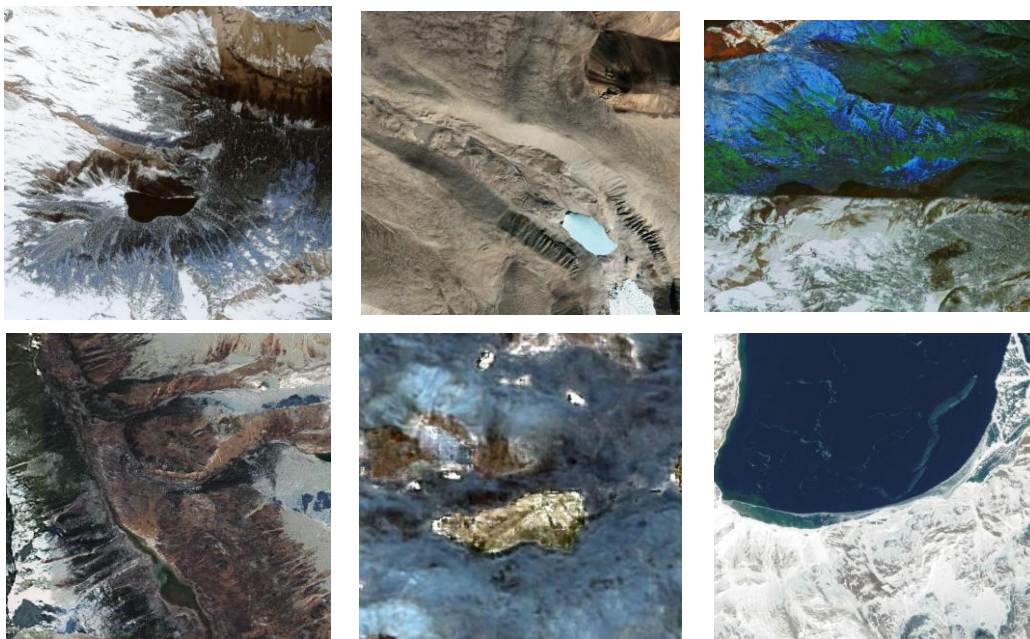


Figure 1 Sample Images

Objectives:

Based on the problem statement given in website and our understanding, the objectives of the challenge are to develop an effective image processing algorithm which is able to:

- Accurately differentiate glacial lakes from highly confounding, spectrally similar environmental features, including seasonal snow cover, clean glacier ice, debris-covered ice, and localized mountain terrain shadows.
- Automatically delineate glacial lake boundaries from high-resolution input satellite imagery to generate precise, pixel-level binary masks.
- Ensure consistent lake identification and tracking across time-series (multi-temporal) satellite observations, maintaining high accuracy despite variations in seasonal illumination, cloud cover, and atmospheric distortion.

Methodology I – Multi-Stage Glacier Lake Segmentation Algorithm (MATLAB)

The methodology consists of a sequence of image processing operations that progressively enhance glacier lake regions while suppressing non-lake regions.

Considerations and Justifications:

Hindrance in Glacier Images	Technique Used	Purpose of the Technique
Cloud Cover and Haze	Contrast Enhancement (CLAHE)	Improves image visibility and highlights lake boundaries

Snow Cover	Snow Masking	Detects and removes bright snow regions that may be mistaken for glacier lakes.
Terrain Shadows	Texture Analysis (Entropy & Variance)	Differentiates dark shadows from water based on surface texture characteristics.
Debris and Moraine Regions	Smooth Region Detection	Identifies smoother lake surfaces while suppressing rough debris and moraine-covered areas.
Turbid or Color-Varying Lakes	Texture-Based Segmentation	Detects lakes using texture properties rather than relying only on water color.
Small and Fragmented Lakes	Morphological Operations	Connects nearby lake pixels, fills gaps, and improves lake continuity.
Noise and False Detections	Region Filtering	Removes unwanted regions that do not satisfy glacier lake characteristics.
Complex Glacier Environment	Multi-Stage Processing Pipeline	Combines enhancement, masking, texture analysis, and filtering to improve overall lake detection accuracy.

Table 1 Justification for each phase in development of MATLAB Algorithm

The detailed procedure is given by:

- i) Acquire the RGB glacier satellite image for analysis.
- ii) Enhance image contrast using CLAHE to improve lake visibility.
- iii) Convert the enhanced image into grayscale, HSV, and LAB color spaces.
- iv) Detect and mask snow-covered regions using brightness and saturation information.

- v) Compute local entropy and standard deviation to analyze texture characteristics.
- vi) Extract smooth regions that potentially represent glacier lakes.
- vii) Generate candidate lake regions by combining texture information and snow masking.
- viii) Apply morphological operations to refine and connect fragmented lake regions.
- ix) Perform connected component analysis to identify individual candidate regions.
- x) Extract region properties such as area, solidity, and mean intensity.
- xi) Filter out non-lake regions using predefined region-based criteria.
- xii) Generate the final binary glacier lake mask.
- xiii) Superimpose the detected lake mask on the original image for visualization.
- xiv) Analyze the final glacier lake detection results.

The above flow is implemented in MATLAB and a sample output is at Figure 2.

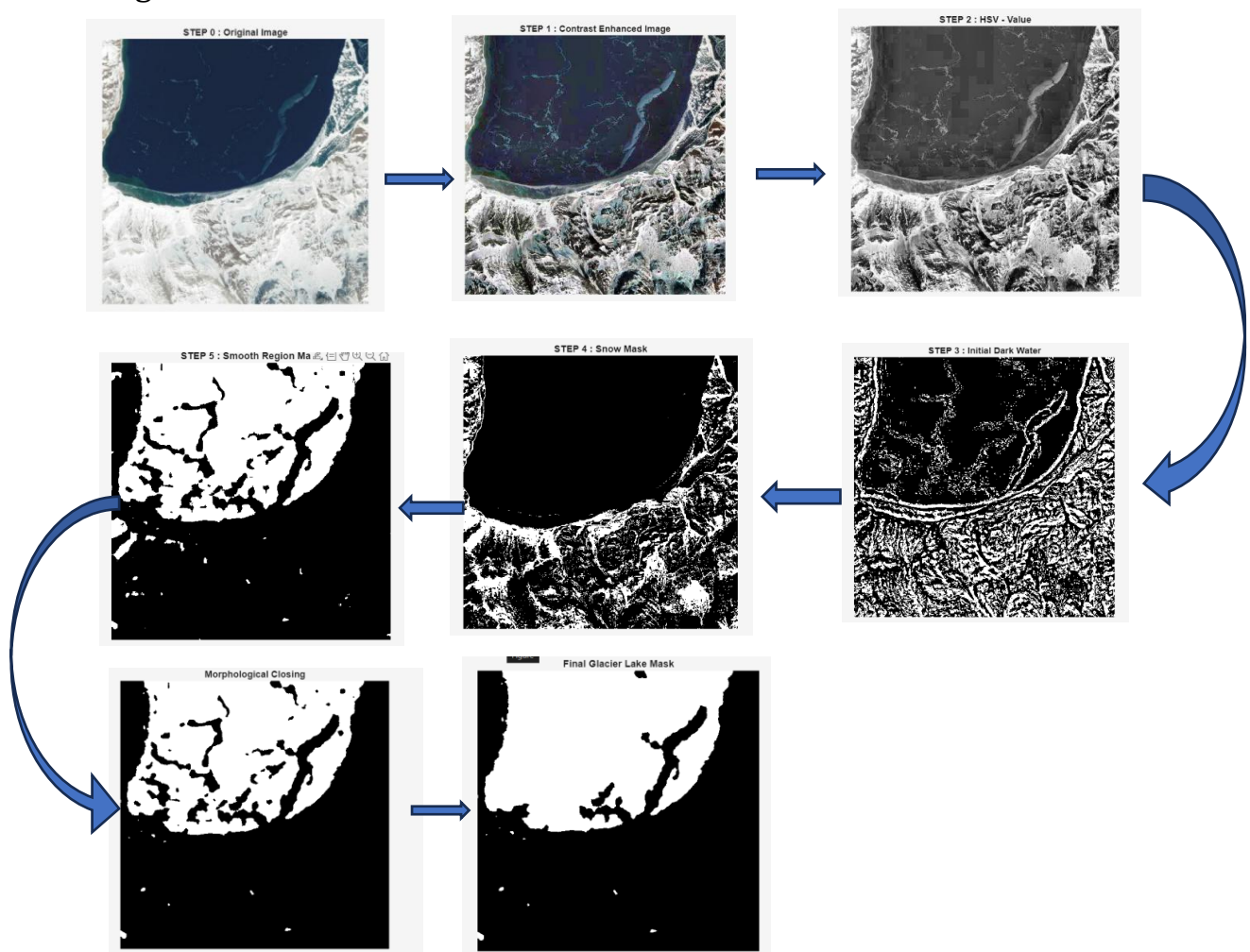


Figure 2 Traditional Image Processing Algorithm Flow

Drawbacks and failing test cases of this methodology:

- i. The algorithm relies on manually selected thresholds and parameters, making it difficult to achieve consistent performance across different glacier environments.
- ii. The method performs poorly in regions with dense moraine cover, as moraine surfaces often resemble glacier lakes in texture and appearance, leading to inaccurate lake boundaries.
- iii. Terrain shadows are frequently misclassified as water bodies because both appear as dark regions in satellite images, causing false detections and mask deviations.
- iv. Glacier lakes exhibit different colors across multi-temporal images due to seasonal variations, illumination changes, and turbidity, reducing the effectiveness of fixed color- and intensity-based techniques.
- v. Small, fragmented, and irregularly shaped lakes may be lost during noise removal and morphological processing, resulting in missed detections.
- vi. Since the method is based on handcrafted image features and lacks learning capability, it cannot adapt to complex glacier conditions, leading to inconsistent and less accurate lake masks.

As the results produced are not promising for many images, evaluation has not been done for this developed image processing algorithm and approach is shifted to deep learning to improve accuracy and handle all the different scenarios so that glacier lake can be segmented.

Methodology II - Iterative Self-Training via Pseudo-Labeling via TransUNet – A Semi-Supervised Learning (SSL) Algorithm

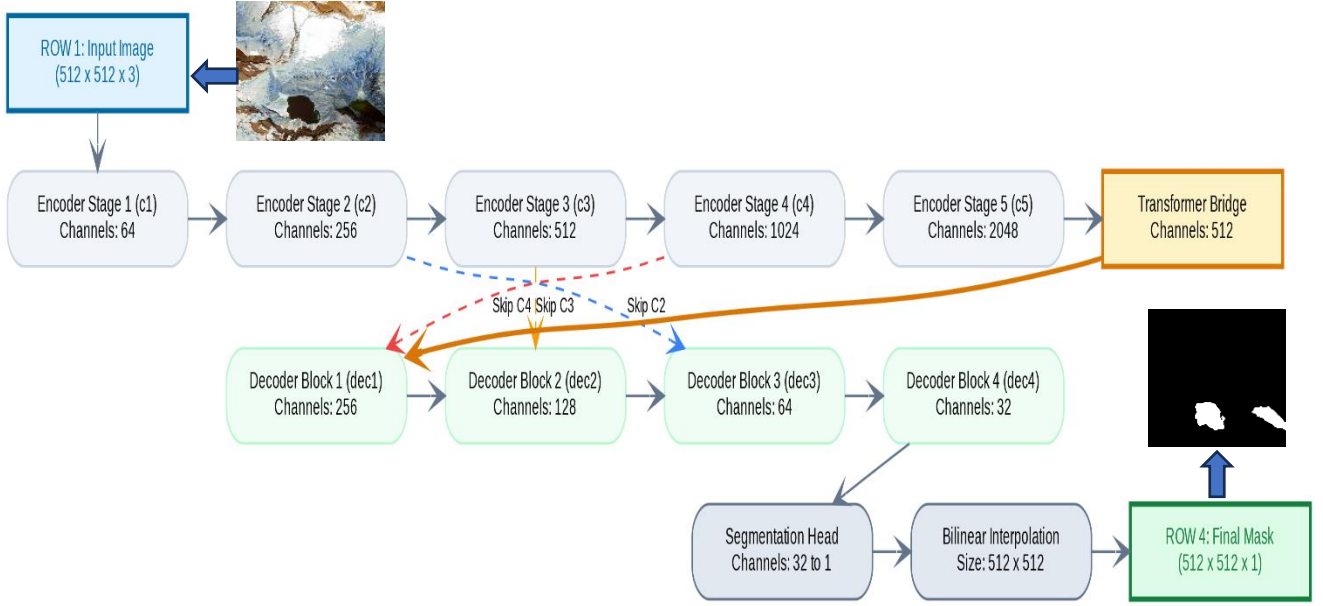


Figure 3 TransUnet Architecture

The methodology leverages an **Iterative Semi-Supervised Self-Training pipeline** driven by a **Hybrid CNN-Transformer architecture (TransUNet)**. Operating under strict hardware and dataset constraints, the framework treats the model as an evolving annotator. The architecture involving the major block diagrams and flow are illustrated above at Figure 3. By deploying a multi-round, progressive pseudo-labeling curriculum, the network automatically scores and extracts high-certainty samples from a massive unannotated image pool over successive training generations. This strategy systematically expands the master training dataset while filtering out out-of-distribution noise and ambiguous edge cases. To stabilize model convergence against the extreme geometric class imbalance between scarce glacial lake pixels and expansive mountain background terrains, the framework optimizes the network using a unified Hybrid Loss objective:

$$\mathcal{L}_{\text{Hybrid}} = \mathcal{L}_{\text{BCE}} + \left(1 - \frac{2 \sum_i P_i Y_i + \epsilon}{\sum_i P_i + \sum_i Y_i + \epsilon} \right)$$

Here, Binary Cross-Entropy $\mathcal{L}_{\text{BCE}}$ ensures stable, pixel-level gradient propagation, while the companion Dice Loss term maximizes global geometric overlap. P_i represents the predicted semantic probabilities, Y_i denotes the corresponding target labels (human annotations or frozen pseudo-labels), and $\epsilon = 10^{-5}$ is a smoothing coefficient. This mathematical safeguard prevents division-by-zero errors during backpropagation across completely water-free satellite frames, ensuring robust training optimization throughout successive active learning generations.

Experimental setup and training

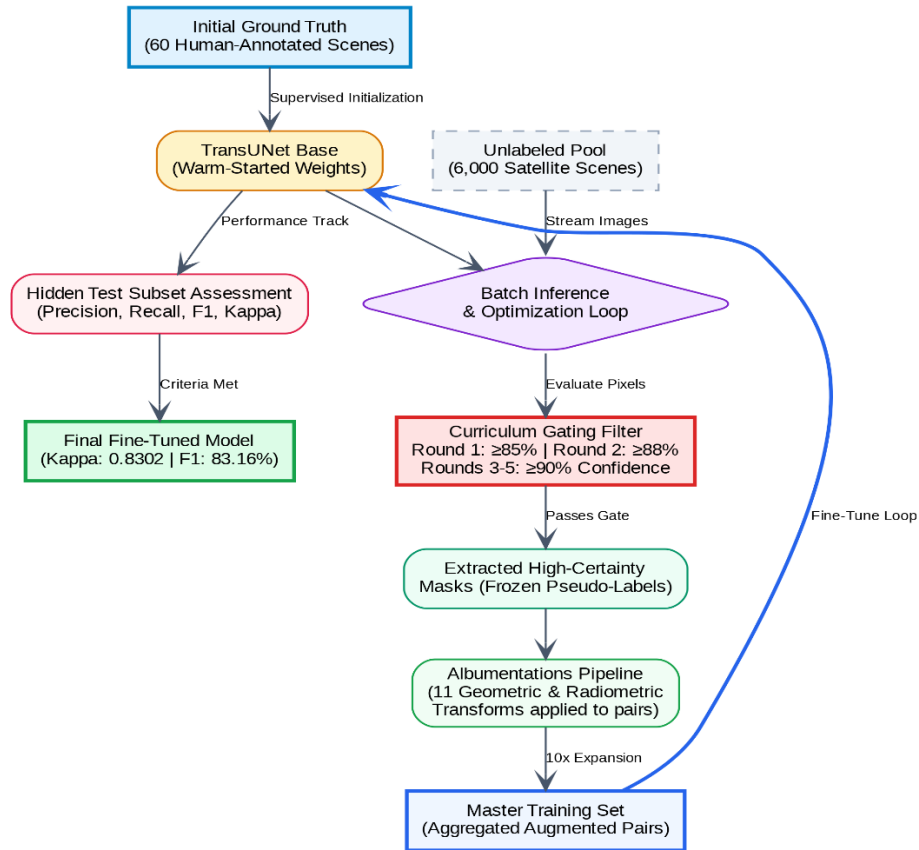


Figure 4 Training methodology Flow

The training phase was executed as a multi-stage, iterative fine-tuning cycle structured to systematically scale model capability while maintaining strict optimization bounds. The training trajectory commenced

with an **Initial Baseline Model** trained for 40 epochs using the provided 60 ground truth labels which were augmented to 660 images(10x times) based on below consideration to establish initial structural parameters on the high-confidence human-annotated dataset.

S.no	Albumentations Transform	Description and purpose
1.	A.HorizontalFlip	Flips the image and mask horizontally
2.	A.VerticalFlip	Flips the image and mask vertically.
3.	A.ShiftScaleRotate	Randomly translates, scales, and rotates scenes
4.	A.HueSaturationValue	Alters color channels to replicate variations in water turbidity and organic depth
5.	A.RandomBrightnessContrast	Modifies illumination to simulate diverse sun elevation angles.
6.	A.GaussNoise	Injects stochastic Gaussian noise into the image to replicate electronic sensor grain and hardware static..
7.	A.GridDistortion	Warps the image cells locally to simulate topographic relief displacement
8.	A.ElasticTransform	Applies localized elastic warping to mimic highly irregular glacial shorelines
9.	A.CoarseDropout	Masks random rectangular image blocks to train the network to segment lakes partially obscured by localized clouds.
10.	A.RandomFog	Simulates alpine mist, haze, and low-altitude cloud cover to test model resilience under degraded visibility.

Table 2 Different Augmentations

Following this initialization, four successive active learning cycles (**Rounds 1 through 4**) were performed. To avoid catastrophic forgetting during these pseudo-label injection stages, the training durations were dynamically compressed from 12 epochs in Round 1 down to a stable 8 epochs by Round 4. The training methodology is represented in figure 4.

The ingestion of unannotated remote sensing instances into the master training dataset was governed by a progressive curriculum gating mechanism. With each round, the model will learn better. So, the confidence threshold τ was strictly scaled across iterations to ensure only premium, high-certainty boundaries were converted into frozen pseudo-labels:

- **Round 1:** $\tau = 85\%$ threshold extracted 100 new scenes.
- **Round 2:** $\tau = 88\%$ threshold extracted an additional 100 scenes.
- **Round 3:** $\tau = 90\%$ threshold swept up 177 complex targets.
- **Round 4:** $\tau = 92\%$ threshold captured the final 103 high-difficulty targets.

In this way, a master augmented dataset of 5940 images with its masks were made and used for 4th round training. After 4th round, 32 residual images from 575 images were not able to cross threshold and on verification, they were found to have no lakes and in some cases the lake areas not determinable. Hence they were left behind in order to prevent the degradation of performance of model.

Additional training details of each round were tabulated as follows. It highlights the gradual increase in size of dataset and confidence gate and reduction in the epoch number to avoid catastrophic forgetting and learn new images better.

Experimental Parameter	Initial Baseline	Active Round 1	Active Round 2	Active Round 3	Active Round 4
Training Epochs	40	12	10	8	8
Batch Size	6	6	4	4	4
Learning Rate	1e-4	1e-5	1e-5	1e-5	1e-5
Confidence Gate (τ)	-	85%	88%	90%	92%
New Unique Scenes Added	60	100	100	177	103
10x Augmentation	600	1000	1000	1770	1030
Total Dataset Size (N)	660	1,760	2,860	4,807	5,940
Final Training Loss	0.2104	0.2637	0.3252	0.3752	0.3946
Final Validation Loss	0.3412	0.3813	0.4003	0.4432	0.4832

Table 3 Training Parameters

Experimental Results and Evaluation

The performance of the fine-tuned TransUNetHybrid model was benchmarked quantitatively across six standard computer vision and statistical tracking metrics. The system achieved a balanced operational baseline across all evaluation criteria, demonstrating that semi-supervised learning loop succeeded in generalizing features across complex satellite landscapes.

Evaluation metric	Formula	Score
Overall Pixel Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	99.69%
Jaccard Index (mIoU)	$\frac{TP}{TP + FP + FN}$	77.20%
Mean precision	$\frac{TP}{TP + FP}$	84.01%
Mean Recall	$\frac{TP}{TP + FN}$	84.08%
Mean F1 Score	$\frac{2 \times Precision \times Recall}{Precision + Recall}$	83.16%
Cohen's Kappa	$\kappa = \frac{P_o - P_e}{1 - P_e}$	0.8302

Table 4 Evaluation Metrics and results

where

- TP = True Positives
- TN = True Negatives

- FP = False Positives
- FN = False Negatives
- P_o = Observed agreement
- P_e = Expected agreement by chance

Conslusions(Quantitative Analysis) drawn from the evaluation metrics are as follows:

- The model achieved an **Overall Pixel Accuracy of 99.69%** . This high value occurs as overall accuracy include TN in the formula. As the given images include more background area than glacier lake area, TN is more and the model is exceptional in finding the backgrounds and thus resulting in high overall accuracy.
- Achieving a Cohen's Kappa of 0.8302 indicates "Almost Perfect Agreement" with ground-truth annotations. This statistically validates that the model's semantic segmentation is highly deliberate, definitively ruling out random guessing despite the extreme land-to-water class imbalance. As a result of high Cohen's kappa,the model is able to detect lakes despite wrong ground truths as follows:

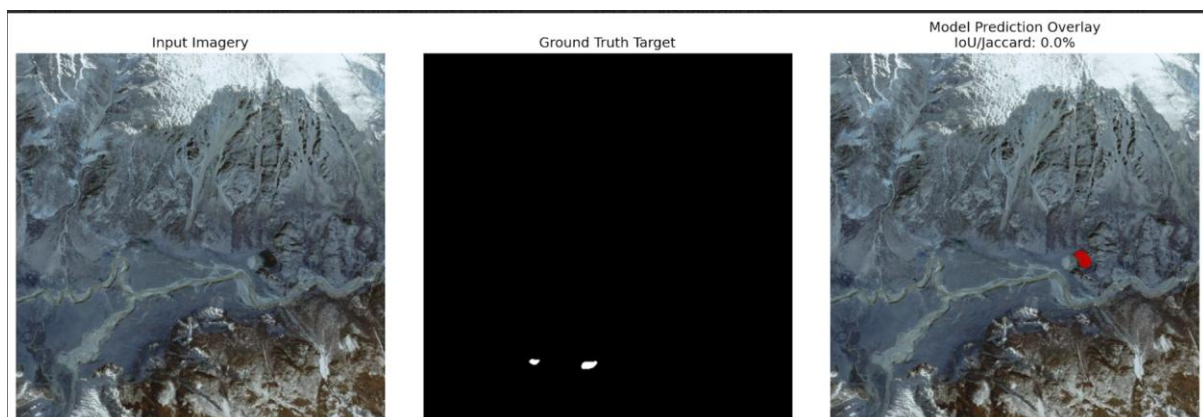


Figure 5 Wrong ground truth 1

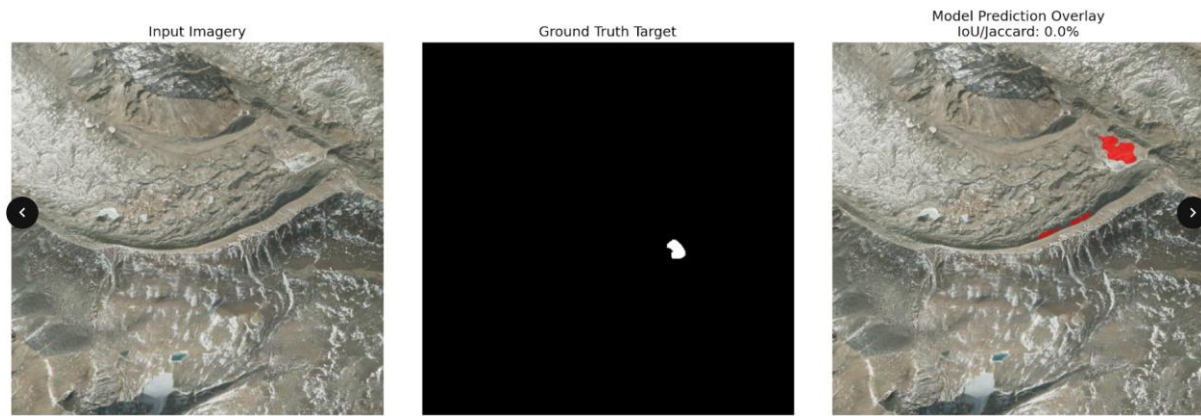


Figure 6 Wrong Ground Truth 2

- **High Precision (84.01%)** proves that the network minimized False Positives (FP).
- **High Recall (84.08%)** demonstrates that the network minimized False Negatives (FN)
- Achieving a Jaccard Index of **77.20%** confirms that the network maintains high geometric fidelity and exact boundary placement on scarce targets.

The edge failing cases were filtered based on low Jaccard Index as follows. They occur due to improper Ground Truth. They are shown above in Figure 4 and 5.

Novelty of the methodology

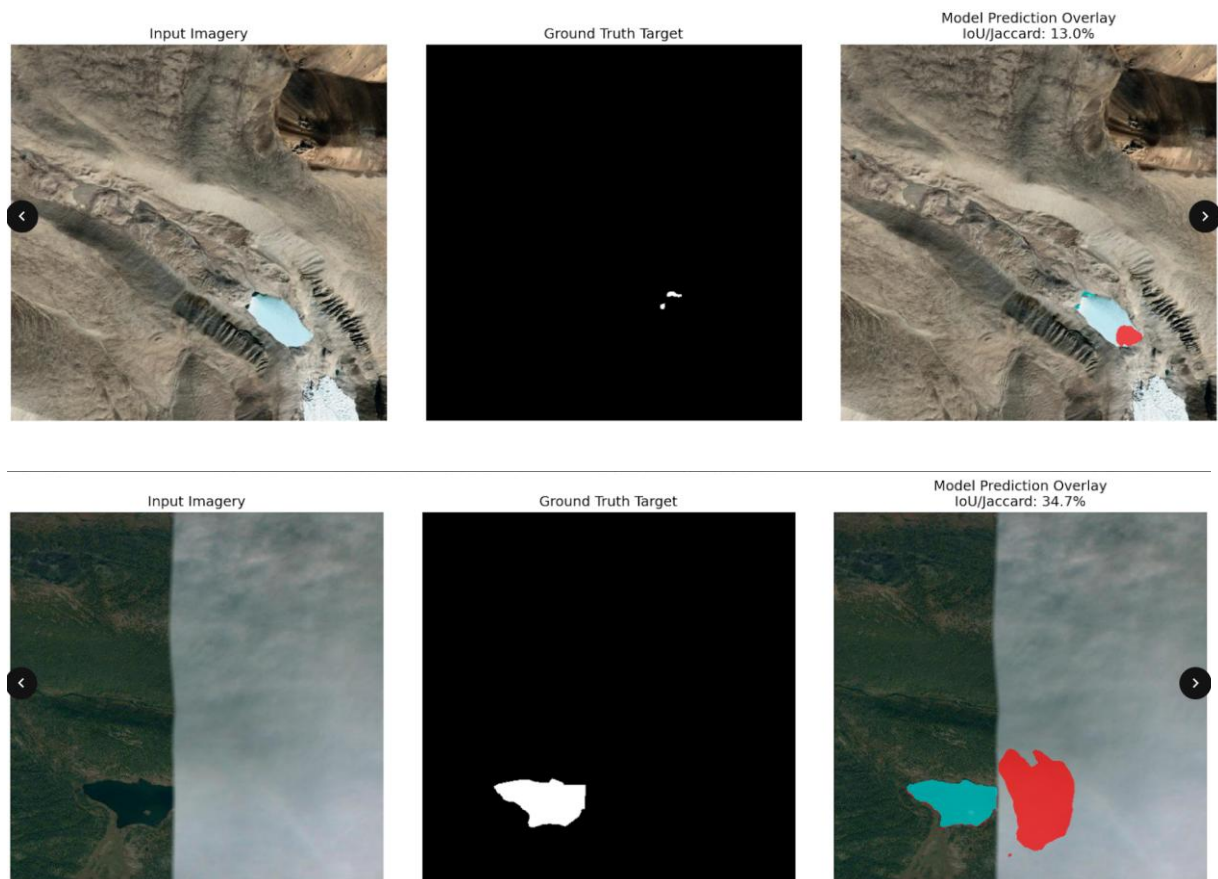
The main novel features of the proposed Deep Learning Algorithm involves

1. Integration of deep ResNet-50 convolutional encoder with a centralized channel-reduction bridge and a cascaded UNet decoder to simultaneously preserve sharp shoreline details and capture long-range global terrain context.

2. Deployment of a multi-round semi-supervised training sequence using progressive confidence gating (85% to 92%) to programmatically scale the operational dataset from 60 human-annotated scenes over 575 images to 5,940 training samples.
3. Utilization of a joint Binary Cross-Entropy and smoothed Dice Loss strategy to mathematically counteract extreme land-to-water pixel imbalances and suppress false-positive terrain shadow segmentations during active training cycle.

Qualitative Analysis

Some failure edge cases which pulled the Jaccard index were due to heavy cloud or glacier lake and normal lake confusions. Almost all images have visibly acceptable lake shape and coverings. The few failed test cases were as follows in Figure 7.



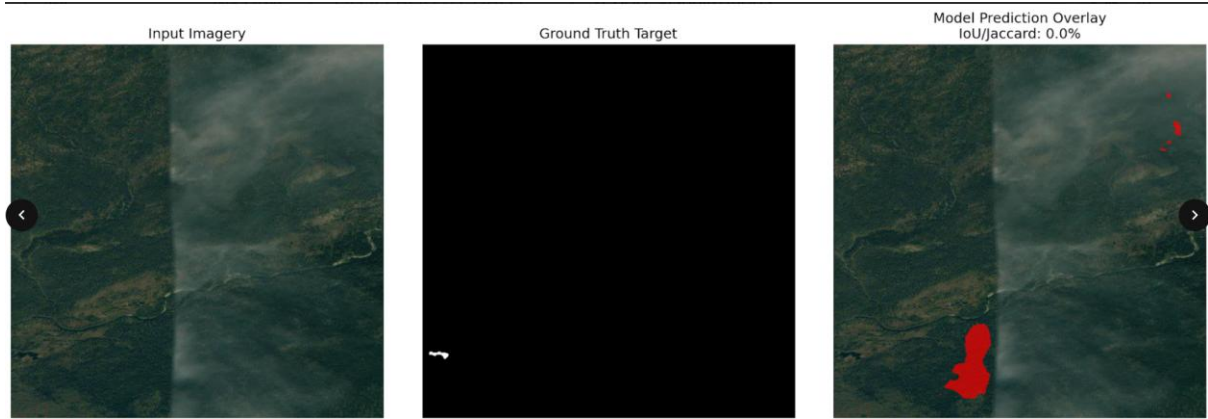


Figure 7 Failed cases

Conclusion

This work successfully engineered and validated a data-efficient, semi-supervised framework for automated glacial lake semantic segmentation across complex alpine topographies. By wrapping a hybrid CNN-Transformer architecture (TransUNetHybrid) inside an iterative, curriculum-driven pseudo-labeling pipeline, the severe constraint of manual data scarcity was bypassed.

Future work will focus on integrating multi-modal data streams, specifically combining optical imagery with Synthetic Aperture Radar (SAR) data from Sentinel-1 to ensure continuous cloud-penetrating monitoring during active monsoon seasons. Additionally, the pipeline will be adapted into a time-series framework to track multi-temporal volumetric lake expansion, providing a highly scalable computational blueprint for real-time Glacial Lake Outburst Flood (GLOF) early warning systems.

References

- [1] J. Chen, Y. Lu, Q. Yu, X. Luo, S. H. Chang, and Y. Wang, "TransUNet: Transformers make strong encoders for medical image segmentation," arXiv preprint arXiv:2102.04306, 2021.
- [2] A. Buslaev, V. Iglovikov, E. Khvedchenya, A. Parinov, M. Druzhinin, and A. A. Kalinin, "Albumentations: Fast and flexible image augmentations," Information, vol. 11, no. 2, p. 125, 2020.

[3] Q. Zhang, L. Zhang, and D. Li, "Semi-supervised semantic segmentation of satellite imagery via iterative progressive pseudo-labeling," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1-14, 2022.

[4] <https://arxiv.org/abs/2102.04306>